

# Statistical Analysis Report

## Analysis Summary Template

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### 1 Summary

Provide a brief overview of the analysis objectives, key findings, and main conclusions. This section should be concise and accessible to non-technical stakeholders. Max 1 paragraph, no bullet points

### 2 Introduction

This analysis is to demonstrate the use of Artificial Intelligence (AI) Large Language Models (LLMs) to analyse data. The goal here is to stress test LLMs on a simple analysis task that involves a linear mixed effects model with a binary dependent variable and a blocking factor. The report will be in the form of a statistical analysis report.

### 3 Data

The data are from a subset of the Chz2006 dataset in the R package {agricolae} (Mendiburu and Yaseen 2020). There are 3 factors of interest (crop, amendment, and block) and a binary dependent variable (wilt\_percent). There is also a plant individual factor which we will not consider here. There are  $n = 1920$  observations in the data. The data experimentally measures the incidence and performance of healthy tubers and rotten potato field infested with naturally *Ralstonia solanacearum* Race 3/Bv 2A, after application of inorganic amendments and a rotation crop in Carhuaz Peru, 2006.

```
'data.frame':  1920 obs. of  6 variables:
 $ amendment  : chr  "OC" "OC" "OC" "OC" ...
 $ crop        : chr  "Pea" "Pea" "Pea" "Pea" ...
```

```

$ block      : Factor w/ 3 levels "1","2","3": 1 1 1 1 1 1 1 1 1 1 1 ...
$ plant      : int   1 2 3 4 5 6 7 8 9 10 ...
$ wilt_percent: int   0 0 50 0 0 0 0 0 0 0 0 ...
$ wilt_bin   : num   0 0 1 0 0 0 0 0 0 0 0 ...

```

Table 1: Dataset sample

| amendment | crop | block | plant | wilt_percent | wilt_bin |
|-----------|------|-------|-------|--------------|----------|
| 0C        | Pea  | 1     | 1     | 0            | 0        |
| 0C        | Pea  | 1     | 2     | 0            | 0        |
| 0C        | Pea  | 1     | 3     | 50           | 1        |
| 0C        | Pea  | 1     | 4     | 0            | 0        |
| 0C        | Pea  | 1     | 5     | 0            | 0        |
| 0C        | Pea  | 1     | 6     | 0            | 0        |

## 4 Statistical Analysis

The statistical analysis consisted of a linear mixed effects model with a binary dependent variable and a blocking factor treating block as a random effect. The model was fitted using the lme4 package in R.

## 5 Results

The results of the analysis are presented below with the predictor names and the corresponding p-values, accompanied by a graph of the relationship between each predictor and the response variable.

```
[1] "Model Summary"
```

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: wilt_bin ~ crop + amendment + (1 | block)
Data: wilt

```

|        |        |        |           |          |
|--------|--------|--------|-----------|----------|
| AIC    | BIC    | logLik | -2*log(L) | df.resid |
| 1146.5 | 1191.0 | -565.3 | 1130.5    | 1912     |

Scaled residuals:

| Min     | 1Q      | Median  | 3Q      | Max    |
|---------|---------|---------|---------|--------|
| -0.9244 | -0.3003 | -0.2264 | -0.0968 | 7.8011 |

Random effects:

| Groups Name       | Variance | Std.Dev. |
|-------------------|----------|----------|
| block (Intercept) | 0.003477 | 0.05897  |

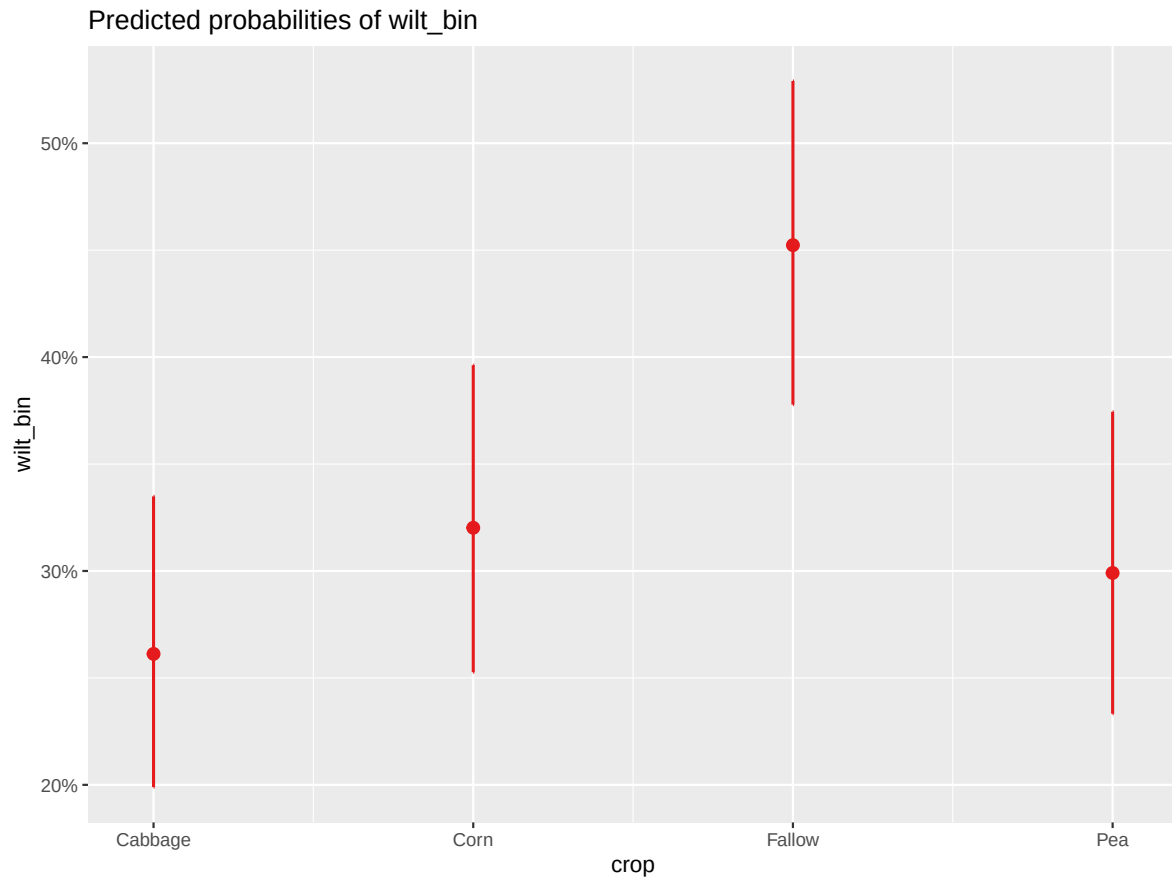
Number of obs: 1920, groups: block, 3

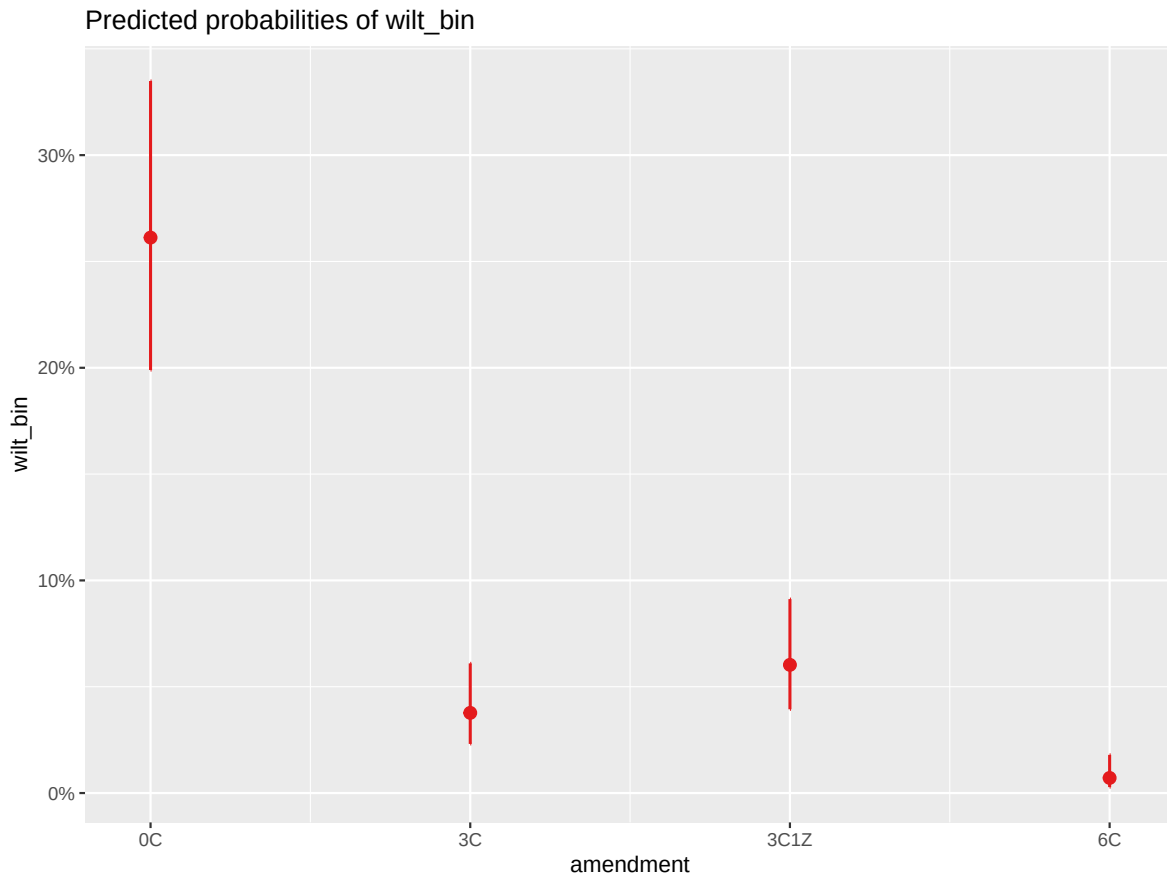
Fixed effects:

|               | Estimate | Std. Error | z value | Pr(> z ) |     |
|---------------|----------|------------|---------|----------|-----|
| (Intercept)   | -1.0396  | 0.1803     | -5.766  | 8.11e-09 | *** |
| cropCorn      | 0.2867   | 0.2291     | 1.252   | 0.211    |     |
| cropFallow    | 0.8484   | 0.2170     | 3.909   | 9.28e-05 | *** |
| cropPea       | 0.1879   | 0.2320     | 0.810   | 0.418    |     |
| amendment3C   | -2.1997  | 0.2254     | -9.761  | < 2e-16  | *** |
| amendment3C1Z | -1.7068  | 0.1917     | -8.905  | < 2e-16  | *** |
| amendment6C   | -3.8991  | 0.4608     | -8.462  | < 2e-16  | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1





## 6 References

Mendiburu, Felipe de and Yaseen, Muhammad(2020). agricolae: Statistical Procedures for Agricultural Research.R package version 1.4.0, <https://myaseen208.github.io/agricolae/><https://cran.r-project.org/package=agricolae>.

## 7 Appendix

### Session Information

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
```

Running under: Windows 11 x64 (build 26100)

Matrix products: default

LAPACK version 3.12.1

locale:

```
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8
```

time zone: Europe/London

tzcode source: internal

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] sjPlot_2.9.0      lmerTest_3.1-3    lme4_1.1-37       Matrix_1.7-3
[5] knitr_1.50        lubridate_1.9.4   forcats_1.0.1     stringr_1.5.1
[9] dplyr_1.1.4       purrr_1.0.4       readr_2.1.5       tidyr_1.3.1
[13] tibble_3.3.0      ggplot2_3.5.2     tidyverse_2.0.0
```

loaded via a namespace (and not attached):

```
[1] generics_0.1.4      stringi_1.8.7      lattice_0.22-7
[4] hms_1.1.3           digest_0.6.37      magrittr_2.0.3
[7] evaluate_1.0.4      grid_4.5.1         timechange_0.3.0
[10] RColorBrewer_1.1-3  fastmap_1.2.0      jsonlite_2.0.0
[13] ggeffects_2.3.1     scales_1.4.0       numDeriv_2016.8-1.1
[16] reformulas_0.4.1    Rdpack_2.6.4       cli_3.6.5
[19] rlang_1.1.6         rbibutils_2.3      splines_4.5.1
[22] withr_3.0.2         yaml_2.3.10        datawizard_1.3.0
[25] tools_4.5.1         tzdb_0.5.0         nloptr_2.2.1
[28] minqa_1.2.8         sjlabelled_1.2.0   boot_1.3-31
[31] vctr_0.6.5          R6_2.6.1           lifecycle_1.0.4
[34] MASS_7.3-65         insight_1.4.2      pkgconfig_2.0.3
[37] pillar_1.10.2       gtable_0.3.6       glue_1.8.0
[40] Rcpp_1.0.14         haven_2.5.5        xfun_0.52
[43] tidyselect_1.2.1    farver_2.1.2       htmltools_0.5.8.1
[46] nlme_3.1-168        labeling_0.4.3     rmarkdown_2.29
[49] compiler_4.5.1
```